Self-Reflective Retrieval-Augmented Generation for Enterprise Anti-Money Laundering Query Systems

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Assignment Notes

Our project satisfies requirement (3) from the assignment description. Specifically, we apply an existing NLP method, Self-RAG, to a new domain: enterprise-level Anti-Money Laundering (AML) tasks. While Self-RAG has been primarily explored in general domains, there is little existing research or implementation guidance on adapting Self-RAG techniques to high-stakes, compliance-driven fields like AML. In this project, we tailor the retrieval and generation processes to address the unique challenges posed by AML data, such as domain-specific terminology, regulatory requirements, and the need for highly factual and auditable outputs. Thus, our project constitutes a novel contribution by adapting an advanced NLP method to a specialized, underexplored application domain.

There is meaningful overlap between our homework 3 and homework 4, but there are also clear differences. In homework 3, we focused on replicating the Self-RAG approach using a wide range of general-domain datasets, closely following the original paper's methodology. In contrast, homework 4 builds on the pre-trained models developed in homework 3, but retrains the retriever component specifically for the AML domain. The model adaptation and retriever fine-tuning reflect the shift toward domain-specific challenges rather than general replication, resulting in a more specialized and applied contribution.

Abstract

Anti-Money Laundering (AML) compliance is a critical but difficult-to-implement task in the finance industry. The regulations can vary depending on the target entity's industry, country, or previous financial activities. This project utilized Self-Reflective Retrieval-Augmented Generation to build an AML domain-specific query system for the financial institutions employees to smooth and accelerate the AML process. Self-RAG addresses RAG limitations by enabling adaptive behavior, on-demand retrieval, and self-evaluation for more accurate and relevant outputs. Reflection tokens enhance output quality by critiquing retrieval relevance, generation support, and utility, ensuring better factual accuracy. For this specific task, we focused on experimenting with various retrieval methods. The results demonstrated that GPT-4 performed best, achieving the highest evaluation scores compared to two other fine-tuned retrievers and the baseline retriever.

1. Introduction

Anti-Money Laundering (AML) compliance is a critical function within financial enterprises, essential for detecting and preventing illicit financial activities. However, accessing and utilizing the vast and often complex body of AML knowledge poses a significant challenge for employees in the KYC (Know Your Client) office in the financial institutions. Analysts and compliance officers frequently spend considerable time manually searching through disparate internal documents, policies, and regulatory guidelines to find answers to specific queries. This inefficiency hinders timely decision-making and increases compliance risk.

To address this challenge, we propose the application of Self-Reflective Retrieval-Augmented Generation (Self-RAG) to build an enterprise-level Question Answering (QA) system for AML. Retrieval-Augmented Generation (RAG) techniques enhance Large Language Models (LLMs) by retrieving relevant information from external knowledge sources before generating a response. Self-RAG extends this by incorporating self-reflection capabilities, allowing the model to dynamically retrieve information and critically evaluate the relevance and quality of retrieved passages and its own generated responses.

Our system enables financial professionals to pose complex AML-related questions in natural language and receive accurate, relevant, and well-grounded answers derived directly from the enterprise's knowledge base. By automating and streamlining information access, this approach aims to improve knowledge accessibility, reduce compliance burdens, and enhance the overall efficiency and effectiveness of AML operations within financial institutions. This paper details the methodology, experimental setup, and evaluation of our Self-RAG system for enterprise AML.

2. Related Work

Enterprise search and Question Answering (QA) systems are crucial for information retrieval in organizations. Traditional keyword-based searches can be limited in handling natural language nuances, often yielding irrelevant results. Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm to enhance Large Language Models (LLMs) by integrating external knowledge. Various RAG architectures exist; basic RAG systems retrieve documents for LLM context, but may introduce noise by retrieving irrelevant information. Ovadia et al. (2023) demonstrated that RAG outperforms fine-tuning for knowledge integration. Gautam & Purwar (2024) showed that Llama3-8B outperforms larger models in enterprise RAG systems and achieves competitive results against proprietary alternatives with lower costs and faster inference times. More advanced architectures, like REALM, aim to improve retrieval accuracy. Models such as Cohere's Command A have also been developed, featuring hybrid attention and strong RAG capabilities (Team Cohere, 2025).

Self-RAG advances RAG by enabling LLMs to adaptively retrieve information and use self-reflection with "reflection tokens" (Asai et al., 2023). This allows dynamic retrieval and critique of outputs, generating more accurate and trustworthy responses. By integrating retrieval, generation, and self-critique, Self-RAG addresses the limitations of indiscriminate retrieval and improves LLM output quality (Asai et al., 2023).

While RAG research is growing, specific applications in Anti-Money Laundering (AML) and financial compliance are relatively new. This paper explores Self-RAG to enhance information retrieval from AML regulatory documents.

3. Methodology

Our approach integrates Self-RAG into a workflow designed for enterprise AML knowledge access. The core components include data preparation, retriever design and fine-tuning, and Self-RAG inference and evaluations.

3.1 Data Acquisition and Preprocessing

The dataset for this study was constructed through web scraping publicly available Anti-Money Laundering (AML) guidance from the Financial Industry Regulatory Authority (FINRA) website (https://www.finra.org/rules-guidance/key-topics/aml/faq)). A total of 232 webpages, encompassing various FAQs and explanatory content related to AML regulations and compliance, were collected and initially stored in a JSON format.

Subsequently, a data cleaning process was implemented to enhance the quality and relevance of the textual data. This involved the normalization of whitespace, including the replacement of newline and tab characters with single spaces, and the removal of redundant multiple spaces using regular expressions. Furthermore, specific patterns indicative of navigational elements, repeated information, and non-content sections (e.g., "Toggle submenu," "Contact Us," website footers) were identified and removed using regular expression matching to isolate the core informational content of each webpage. Instances of ellipsis were standardized to periods for consistency in sentence segmentation.

Given the presence of non-English text within the scraped data, likely due to embedded content or interface elements, a translation step was performed using the GPT-4o model via API to ensure uniformity and facilitate downstream processing in English. The structured JSON format of the processed data, as illustrated in Figure 1, retained key information such as the source URL and the main textual content of each webpage.

```
[
  {
    "url": "https://www.fdic.gov/",
    "type": "HTML",
    "level": 4,
    "content": "FDIC: Federal Deposit Insurance Corporation Skip to main content An official website
    of the United States government The.gov means it’s official. Federal government websites often
    end in.gov or.mil. Before sharing sensitive information, make sure you’re on a federal
    government site. The site is secure. The https:// ensures that you are connecting to the
    official website and that any information you provide is encrypted and transmitted securely.
    Cambiar a español   About  About The Federal Deposit Insurance Corporation (FDIC) is an
    independent agency created by the Congress to maintain stability and public confidence in the
    nation’s financial system. Learn about the FDIC’s mission, leadership, history, career
```


Figure 1. Format of data scraped

For subsequent analysis and the creation of Question-Answer (QA) pairs, the content of each webpage was segmented into smaller, manageable chunks using a maximum token length of 300 words. This chunking strategy aimed to maintain contextual coherence while accommodating the input constraints of the QA generation model.

Finally, following the ASQA style, a total of 600 Question-Answer pairs were generated using the GPT-4o API. For each content chunk, three distinct, enterprise-relevant AML questions were formulated, along with their corresponding grounded and factual long-form answers, a list of 2-3 concise short answers, a 1-2 sentence contextual summary, and the specific supporting text extracted from the content chunk. The prompt used to guide the GPT-4o model in this QA pair generation process is provided below:

PROMPT

```
You are a financial compliance assistant
helping to build a Self-RAG QA dataset for
enterprise AML education. Your task is to
generate 3 related question-answer pairs
from the topic below.

Each QA pair must include:
- `question`: enterprise-relevant AML question
- `answer`: grounded, factual, full answer
- `short_answers`: list of 2-3 concise answers
- `context`: (1-2 sentence summary to support
or frame the question)
- `source_text`: supporting content from the
passage

Rules:
- Use ONLY the provided content as source
- Questions must be distinct, topically related
- Do NOT repeat or paraphrase the same
question
- All fields must be completed per QA pair
```

FORMAT

```
Return a valid JSON list like:
[
 {
 "question": "...",
 "answer": "...",
 "short_answers": ["...", "..."],
 "context": "...",
 "source_text": "..."
 },
 ...
]

---CONTENT START---
{topic['content']}
---CONTENT END---
```
```

*Figure 2. Figure of prompt and qa_pais's format*

This structured dataset of QA pairs forms the basis for subsequent stages of this research.
```

3.2 Retriever Design and Fine-tuning

A key component of the RAG system is the retriever, responsible for fetching relevant document passages given a query. We experimented with different retriever configurations:

1. Default Retriever: Utilized the pre-trained `facebook/contriever` model. An initial retrieval database (FAISS index and metadata) was built using embeddings from this model.
2. Fine-tuned Retriever: To potentially improve retrieval relevance for the specific AML domain, we fine-tuned the `facebook/contriever` model. Our goal was to maximize the similarity between the query and positive docs, and minimize the similarity between the query and the negative docs. The FAISS library was used for efficient vector similarity search in the retrieval database.
    - Retriever Fine-tuning Parameters
        i. `learning_rate`: 2e-5
        ii. `batch_size`: 8
        iii. `num_epochs`: 3
        iv. `temperature`: 0.05
    - After fine-tuning, the retrieval index was rebuilt using the fine-tuned model embeddings.
3. GPT-4o Pre-retrieval: As a point of comparison, we also evaluated using GPT-4o to pre-retrieve relevant documents, leveraging its strong understanding capabilities. The logic for generating the retrieved documents is also included in the prompts above. GPT-4o will automatically provide chunks for their qa_paris resource.

3.3 Self-RAG Inference

We employed the Self-RAG model (specifically selfrag_llama2_7b as per README inference commands) for generation. The Self-RAG process involves iterative retrieval, generation, and critique. Given a query, the model retrieves relevant passages using the configured retriever (default or fine-tuned). It then generates response segments, evaluating them using reflection tokens that assess relevance, grounding (supported by retrieved passages), and utility. This self-critique guides the generation process, potentially involving further retrieval steps or refining the answer based on evidence.

Key inference parameters explored include:

- ndocs: Number of documents retrieved (e.g., 4 or 5).
- max_new_tokens: Maximum length of the generated answer (e.g., 300 or 400).
- threshold: Relevance threshold for retrieval decision (e.g., 0.2).
- max_depth: Maximum depth for the beam search or tree search during generation (e.g., 7).
- Reflection tokens: use_grounding, use_utility, use_seqscore were enabled.
- Retrieval mode: always_retrieve was used in the examples.

(Note: Mention the specific Self-RAG model variant used, e.g., selfrag_llama2_7b, based on the README. Clarify the 'task' parameter set to 'asqa'.)

4. Experiments

4.1 Experimental Setup

- Dataset: Describe the test dataset used for evaluation (derived from full_qa_dataset.json mentioned in the README?). Specify the number of test QA pairs.
- Baselines: Clearly define the baseline(s). The 'Default retriever' configuration likely serves as one baseline. Was there a non-RAG baseline (e.g., Llama 2 7B without retrieval)?
- Configurations: List the main configurations tested, referencing the poster and README:
  - Baseline: Default Contriever + Self-RAG (ndocs=5, max_new_tokens=300, threshold=0.2, max_depth=7)
  - Fine-tuned Retriever + Self-RAG (Simple Prompt? ndocs=4, max_new_tokens=400, threshold=0.2, max_depth=7)
  - Fine-tuned Retriever + Self-RAG (Detailed Prompt? ndocs=5, max_new_tokens=300, threshold=0.2, max_depth=7)
  - GPT-4 Pre-retrieval + Self-RAG (ndocs=?, max_new_tokens=300, threshold=0.2, max_depth=?) *(Need to clarify ndocs and max_depth for GPT-4 setup from poster table). (Clarify what "Simple Prompt" vs "Detailed Prompt" refers to - was this the prompt given to the Self-RAG model during inference, or related to the QA pair format?)*

4.2 Evaluation Metrics

- Answer Correctness: QA-F1, QA-EM
- Retrieval Quality: STR-EM, STR-Hit
- Text Quality: ROUGE-Lsum, MAUVE
- LLM (GPT-40-mini) based Metrics: Factual correctness, completeness, relevance, retrieval utilization
- Other: Answer Length, QA-Hit

5. Results

The experimental results provide valuable insights into the effectiveness of different retrieval strategies when combined with the Self-RAG model for the Anti-Money Laundering (AML) Question Answering task. We compare **four configurations**: a baseline using the default Contriever retriever, two variations using a fine-tuned Contriever retriever (differing slightly in parameters and potentially prompt style), and one using GPT-4o for pre-retrieval. **Evaluation** uses both standard NLP metrics and LLM-based judgments (via GPT-4o-mini).

5.1 GPT-4 Pre-retrieval Performance:

The configuration leveraging GPT-4 for pre-retrieval stands out as the top performer across most metrics related to factual accuracy and relevance.It achieves the highest ROUGE-Lsum (42.52), indicating the generated answers have the greatest content overlap with the reference answers.It significantly leads in answer correctness metrics QA-EM (1.2) and QA-F1 (14.36), suggesting the documents retrieved by GPT-4o were most likely to contain the information needed for an accurate answer.In the LLM evaluation, it scores highest in Compliance Coverage (3.53/5.00), Source Utilization (2.87/5.00), Citation Accuracy (2.95/5.00), and consequently, the Overall Score (3.42/5.00). It also ties for the highest Regulatory Accuracy (4.34/5.00).However, it scores lowest on MAUVE (8.32), suggesting that while factually superior, its generated text might be perceived as less natural or fluent compared to the other configurations. Its answer length is moderate (29.86)**.**

5.2 Fine-tuned Retriever Performance:

Fine-tuning the Contriever retriever demonstrates clear benefits over the default baseline, validating the domain adaptation effort. Both fine-tuned configurations achieve significantly higher ROUGE-Lsum scores (~37.5) compared to the baseline (35.3), indicating improved retrieval of relevant content.They show a modest improvement in QA-F1 (~11) over the baseline (9.22). However, their QA-EM (0.24) is lower than the baseline's (0.6), suggesting they might retrieve generally relevant context but struggle with pinpointing exact answer phrases compared to the baseline in some cases. Notably, the fine-tuned configurations achieve the highest MAUVE scores (9.64 and 9.32), suggesting potentially more fluent or natural text generation compared to both the baseline and GPT-4 pre-retrieval. In the LLM evaluation, the fine-tuned model matches GPT-4o's Regulatory Accuracy (4.34/5.00) and outperforms the baseline. However, it lags behind GPT-4o and sometimes the baseline in other LLM metrics like Compliance Coverage (3.09) and Source Utilization (2.01), resulting in an Overall Score (3.08) slightly above the baseline (3.06) but below GPT-4o.The fine-tuned models produce shorter answers on average (26.4 and 23.49) compared to the others

5.3 Baseline Default Retriever Performance:

The baseline configuration provides a reference point.It has the lowest ROUGE-Lsum (35.3) and QA-F1 (9.22), indicating weaker performance in retrieving relevant content and generating factually overlapping answers compared to the adapted models. Its MAUVE score (9.2) is decent, better than GPT-4 but slightly lower than the fine-tuned models. Its LLM evaluation scores are generally the lowest, particularly in Regulatory Accuracy (4.01), although it surprisingly scores slightly better than the fine-tuned model on Compliance Coverage (3.25) and Source Utilization (2.28) in this specific evaluation run. It generates the longest answers on average (32.99).

5.4. Key Observations and Analysis:

Accuracy vs. Fluency Trade-off: There's a clear trade-off. GPT-4o pre-retrieval maximizes accuracy and relevance (ROUGE, QA metrics, LLM scores) but sacrifices some fluency (MAUVE). Fine-tuned Contriever enhances fluency (MAUVE) and improves relevance over the baseline (ROUGE) but doesn't reach GPT-4's factual accuracy levels.

Value of Fine-tuning: Domain-specific fine-tuning demonstrably improves the retriever's ability to find relevant content (ROUGE increase) compared to the default Contriever.

LLM Evaluation Nuance: The LLM-based metrics provide a more holistic view. While GPT-4o wins overall, the fine-tuned model's ability to match its regulatory accuracy is significant. The relatively low Source Utilization scores across all models suggest challenges in effectively integrating and citing retrieved information, a potential area for future improvement in the Self-RAG generation process itself.

Low QA-EM: The very low Exact Match scores across all configurations highlight the difficulty of the task, suggesting that answers often require synthesis or paraphrasing rather than simple extraction.

The choice of retriever presents a trade-off. For applications prioritizing maximal factual accuracy and compliance coverage in the AML domain, leveraging a powerful model like GPT-4o for pre-retrieval appears most effective, despite potential costs and lower MAUVE scores. However, fine-tuning a dedicated retriever like Contriever offers a viable, potentially more efficient alternative that improves significantly over a generic baseline and can even excel in text fluency, while achieving competitive regulatory accuracy according to LLM evaluation.

	fine-tuned retriever + simple prompt --ndocs 4 --max_new_tokens 400	**fine-tuned retriever + detailed prompt --ndocs 5 --max_new_tokens 300**	**GPT 4o pre-retrieval**	**Baseline**
		NLP Evaluation Metrics HW3 Paper		
length	26.4	23.49	29.86	32.99
rougeLsum	37.56	37.45	**42.52**	35.3
QA-EM	0.24	0.24	**1.2**	0.6
QA-F1	10.8	11	**14.36**	9.22
mauve	9.64	**9.32**	8.32	9.2
		LLM Evaluation Results (with GPT-4o-mini)		
regulatory_accuracy	-----	**4.34/5.00**	**4.34/5.00**	4.01/5.00
compliance_coverage	-----	3.09/5.00	**3.53/5.00**	3.25/5.00
source_utilization	-----	2.01/5.00	**2.87/5.00**	2.28/5.00
citation_accuracy	-----	2.86/5.00	**2.95/5.00**	2.70/5.00
overall_score	-----	3.08/5.00	**3.42/5.00**	3.06/5.00

6. Discussion and Future Work

Our results indicate that the choice of retriever significantly impacts the performance of the Self-RAG system in the AML domain. While leveraging a powerful LLM like GPT-4 for pre-retrieval yielded the best results in terms of answer correctness (QA-F1/EM) and relevance (ROUGE, STR metrics), it may come at the cost of text fluency (lower MAUVE) and potentially higher computational cost or API reliance.

Fine-tuning the Contriever model provided a modest improvement over the default baseline, particularly in ROUGE-Lsum, suggesting that domain-specific tuning is beneficial but may require further optimization or more extensive training data. The discrepancy between higher MAUVE scores and lower QA scores for the Contriever-based methods warrants further investigation.

Based on these findings and observations during development, potential improvements include:

- Prompt Engineering: Adding few-shot examples to the inference prompt to better guide the Self-RAG model's output style and structure.
- Parameter Tuning: Experimenting with retrieval parameters, such as increasing ndocs (e.g., to 10) to provide more context, potentially combined with adjusting max_depth (e.g., to 3-5) to manage generation complexity. *(Need to explain the rationale for decreasing max_depth if increasing ndocs)*.
- Retriever Enhancement: Continuing the fine-tuning process for Contriever, ideally using more high-quality, domain-specific AML Q&A pairs. Exploring hybrid retrieval strategies, perhaps using an initial retriever like Contriever followed by a re-ranker (potentially LLM-based), could balance efficiency and quality.

The findings suggest that while powerful LLMs excel at retrieval via prompting, structured RAG systems like Self-RAG offer advantages in terms of traceability, control, and integration into enterprise workflows. Hybrid approaches, potentially using LLMs as components within a RAG architecture (e.g., as retriever, re-ranker, or critique mechanism), appear promising for optimizing the trade-offs between cost, relevance, and fluency in specialized domains like AML.

7. Conclusion

This paper presented the design and evaluation of a Self-RAG system tailored for Anti-Money Laundering question answering within an enterprise context. We demonstrated the feasibility of using Self-RAG to provide natural language access to complex AML knowledge. Our experiments compared different retrieval strategies, highlighting the strong performance of LLM-based pre-retrieval (GPT-4o) and the potential benefits of fine-tuning dedicated retrievers like Contriever. While GPT-4o pre-retrieval led to higher factual accuracy, fine-tuned models showed competitive performance with potentially better fluency. Our work suggests that hybrid RAG architectures, integrating the strengths of different retrieval methods and leveraging Self-RAG's reflective capabilities, offer a promising direction for building effective, controllable, and efficient enterprise QA systems in regulated domains like AML.

8. References

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